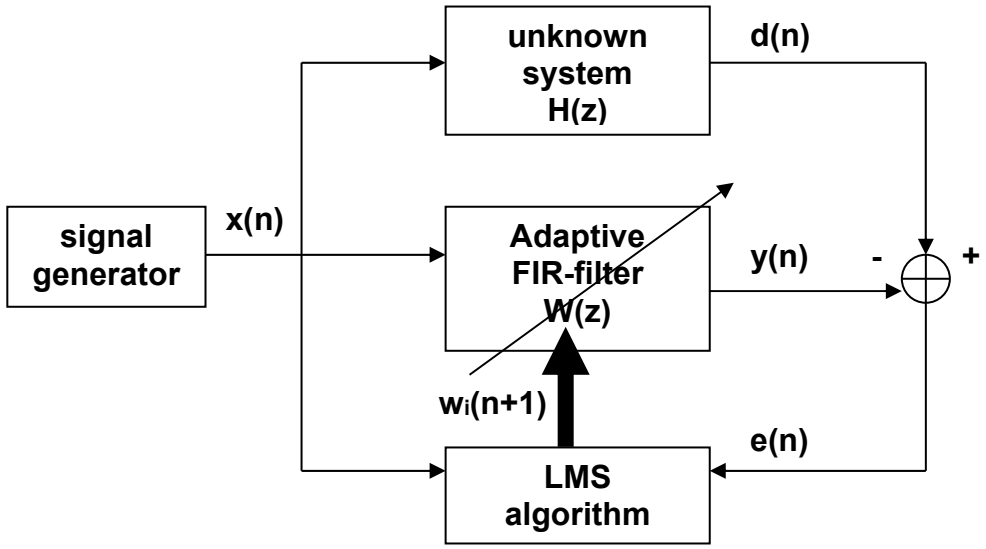
Problem 11.1 (MATLAB)
The LMS algorithm and Adaptive System Identification

In this problem is studied the standard LMS algorithm and its variants applied for adaptive system identification.
The block diagram for adaptive system identification is shown below:



System set-up

- Define an "unknown" (discrete-time) FIR LTI-system, **H(z)**, in MATLAB
  Start with a length of **h(n)** like, e.g., $\mathbf{L_h = 25}$
- Define an input signal **x(n)**, start, e.g., with a zero-mean stationary Gaussian white noise signal of length **N = 1000**
- Generate the "desired" signal, **d(n)**, by filtering **x(n)** with **H(z)**

- Implement the standard LMS algorithm as a function m-file or use the implementation made available accompanying this problem

- Defining first a length of the adaptive filter, $\mathbf{L = L_h}$ and a step size, **μ**, approximately two decades below the stability limit, use the chosen LMS implementation to iterate through the **N** samples in accordance with the block diagram
- Plot the signals **x(n)**, **d(n)**, **y(n)**, **e(n)**
- Visualize the convergence of the algorithm by calculating and plotting the Average Squared Error, $\mathbf{ASE(n)} = \frac{\mathbf{1}}{\mathbf{K}}\sum_{\mathbf{m=n-(K-1)}}^{\mathbf{n}} \mathbf{e^2(m)}$ with, e.g. **K = 5L** or **K=10**
- Plot also the true filter coefficients of **h(n)** along with the filter coefficients of the adaptive filter, **w(n)**, and the true amplitude characteristic **|H(f)|** along with **|W(f)|**

Experiments

- Increase gradually the step size, **μ**, up to approximately the stability limit and observe plots in terms of convergence speed, stationary error and achieved estimate of **H(f)**
- Experiment with **W(z)** lengths, **L**, bigger and smaller than **H(z)** length $\mathbf{L_h}$ and observe plots
- Substitute the FIR **H(z)** with an IIR **H(z)** and view plots
- Optionally, let **x(n)** be low pass filtered white noise – what happens ?

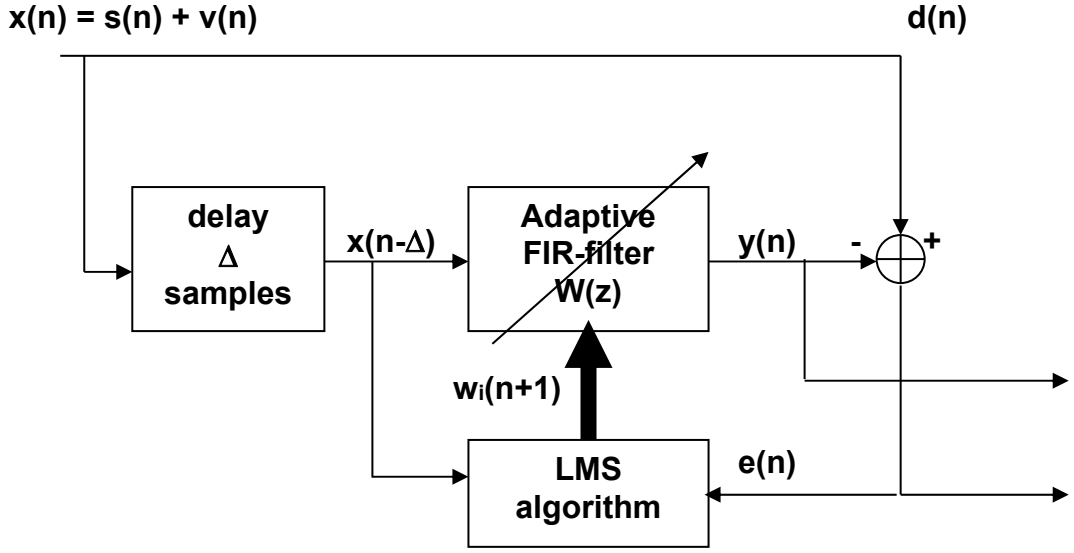
- Now, for the following, implement the Leaky Normalized LMS algorithm or use the implementation made available

- Use the LNLMS algorithm for some of the experiments above and plot now also the time-varying step size, **μ(n)**

- To motivate using the LNLMS instead of the standard LMS, introduce non-stationarity in the block diagram system, e.g. by making **x(n)** non-stationary Gaussian white noise (i.e. with time-varying power)
- Plot again the time-varying step size, **μ(n)**
- Optionally, try to make the unknown system **H(z)** time-varying also

Problem 11.2 (MATLAB)
The LMS algorithm and Adaptive Linear Prediction / Noise Reduction

In this problem is studied the LNLMS algorithm applied for linear prediction and noise reduction.
The block diagram for adaptive linear prediction is shown below:



System set-up

- Define a length **N** narrowband signal **s(n)**, e.g. a single sinusoid or several close sinusoids
- Define a length **N** wideband noise signal **v(n)**, e.g. zero-mean white Gaussian noise
- Let the input signal be **x(n) = s(n) + v(n)**

- Define a delay of **Δ** samples, with **Δ** big enough to have zero correlation between **v(n)** samples more than **Δ** samples apart (if **v(n)** is white noise, **Δ = 1**)
- Generate **x(n- Δ)** by introducing **Δ** leading zeroes in **x(n)**

- Use the LMS or LNLMS algorithm from Problem 11.2 to iterate through the **N** samples in accordance with the block diagram

Experiments

- Start with a relatively high SNR between **s(n)** power and **v(n)** power, a filter length of e.g. **L = 100** and a step size, **μ**, approximately two decades below the stability limit
- Plot the signals **s(n)**, **v(n)**, **x(n)**, **y(n)**, **e(n)**
- Visualize the convergence of the algorithm by calculating and plotting the Average Squared Error, $\mathbf{ASE(n)} = \frac{\mathbf{1}}{\mathbf{K}}\sum_{\mathbf{m=n-(K-1)}}^{\mathbf{n}} \mathbf{e^2(m)}$ with, e.g. **K = 5L** or **K=10**
- Plot also the amplitude characteristic **|W(f)|**
- Calculate the SNR for **y(n)** and compare with the SNR for **x(n)**

- Experiment with different SNR's for **x(n)**
- Experiment with different filter lengths, **L**
- Experiment with different step sizes, **μ**